A First Lean Formalization Audit of the IUT III Corollary 3.12 Corridor

IUT Lean formalization project

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Abstract

This note records the present state of a Lean formalization of a narrow part of Mochizuki's inter-universal Teichmuller theory: the passage from IUT III, Theorem 3.11 to Corollary 3.12, with attention to the Scholze–Stix objection. The current Lean code proves conditional route theorems and obstruction lemmas. It does not prove Corollary 3.12 and does not refute it. The main formal result so far is that a nonarchimedean (Ind3) entry, together with explicit Hodge/SHE square-weight transport data, feeds a hull/log-volume q-to-Θ comparison, while several simplified collapse mechanisms are rejected by Lean.

1 Scope

The review used the local Markdown conversions of IUT I–IV, Mochizuki's 2026 formalization note, and the Scholze–Stix note. The relevant source passages are:

- IUT I: initial Θ-data, prime strips, Hodge theaters, and non-scheme-theoretic Θ-links.
- IUT II: Hodge–Arakelov evaluation, theta values of shape q^{j^2}, multiradiality, and Gaussian monoids.
- IUT III: Theorem 3.11; Corollary 3.12, especially Steps (x) and (xi); Remarks 3.12.2 and 3.12.3.
- IUT IV: later explicit log-volume estimates and Diophantine consequences.
- Scholze–Stix: the ordered real-line identification problem in their Section 2.2.
- Mochizuki 2026: the "3.11.5 $\Rightarrow$ 3.12" fourth-triangle decomposition.

2 Mathematical Target

IUT III Step (x) treats procession-normalized log-volumes as invariants for (Ind1) and (Ind2), and turns (Ind3) into an upper inequality. Step (xi) then uses the multiradial construction, IPL, SHE, holomorphic hulls, determinants, and log-volume to place the q-pilot log-volume in an upper ray controlled by the Θ-pilot log-volume:

$$-|\log(q)| \in \mathbb{R}_{\leq -|\log(\Theta)|} \quad \Rightarrow \quad -|\log(q)| \leq -|\log(\Theta)|.$$

Mochizuki's 2026 note isolates the final comparison as:

$$\text{Theorem 3.11} \xrightarrow{\text{APT+IPL}} \text{“3.11.5”} \xrightarrow{\text{SHE+IPL}} \text{Corollary 3.12.}$$

The present formalization targets only this corridor, not the full construction of Theorem 3.11.

3 Dispute Interface

Scholze–Stix require explicit bookkeeping for all ordered one-dimensional real vector spaces. Their pressure point is that the Θ-side concrete objects carry j^2-scaled degrees, while a consistent identification of the real lines appears to leave no room for inserting all these scalars.

The formalization therefore separates:

$$\begin{array}{c} \text{representative } j^2 \text{ data} \\ \text{balanced sign-compatible data} \\ \text{aggregate averaged data} \\ \text{final hull/log-volume data} \end{array} \quad \text{and} \quad \begin{array}{c} \text{raw real equality} \\ \text{transported equality} \\ \text{matched transport scales} \\ \text{cancellable ordered-line comparison.} \end{array}$$

This separation is the main correctness guard in the current code.

4 Lean Coverage

The relevant implementation is in

`Iut/Stage1/IUTStage1Source.lean` and `Iut/Stage1/IUTStage1Experiments.lean`.

The code currently covers:

- labeled real-line copies and positive-scale transports;
- a finite $\mathbb{F}_\ell$-label model with representative square weights $j \mapsto j^2$;
- a representative pilot-degree profile $\deg(\Theta_j) = j^2 \deg(q)$;
- proof that one label-independent scale cannot absorb all representative j^2 scales in this model;
- proof that, for nonzero $\deg(q)$, one label-independent scale cannot account for all representative Θ_j-degrees;
- proof that the raw representative j^2 theta-degree rule does not descend to the sign quotient $|F_\ell| = F_\ell/\{\pm 1\}$ when $\deg(q) \neq 0$;
- an absolute-label exponent on $|F_\ell| = \{0\} \cup F_\ell^\times/\{\pm 1\}$ which agrees with j^2 for the canonical representatives $0 \leq j \leq (\ell - 1)/2$, and the corresponding theta-degree profile;
- the degree-level Gaussian evaluation $q \mapsto (q^{j^2})_{j \in |F_\ell|}$, including identity at $j = 1$;
- a log-volume shadow of the IUT III splitting-monoid action: the generator attached to a procession label contributes $j^2 \deg(q)$, contributes 0 at the core label, and adds its finite generator sum to the tensor-packet log-volume;
- the finite procession containers $S^\pm_{j+1} = \{0, \ldots, j\}$, their nested inclusions, core-label stability, stage cardinality $j + 1$, and total procession ambiguity $(\ell^\pm)!$ rather than $(\ell^\pm)^{\ell^\pm}$;
- the map from the final procession container to $|F_\ell|$, sending the core label to 0 and recovering the exponent j^2;
- the log-volume shadow of the tensor packet over $S^\pm_{j+1}$, with each factor a finite-fiber direct sum over places above $v_{\mathbb{Q}}$, total log-volume given by the finite sum over the container, normalized denominator $j + 1$, and invariance under container permutations;

- base-valuation products of tensor packets, the finite $(F^{\times}_{\mathrm{mod}})_j$-action shadow, and a coric transfer shadow $F \to D \to D^{\vdash}$ preserving product log-volume without identifying Hodge-theater histories;
- the Step (xi-d)–(xi-g) shadow: determinant normalization, upper-ray membership, and equality of the two q-pilot log-volume computations;
- the Step (xi-h) warning: for $N \geq 2$ and $\Theta < 0$, the tensor-power boundary satisfies $N\Theta < \Theta$;
- the Step (xii) shadow: local Frobenioid p_v^N-ambiguity changes log-volume by an integer shift, while global calibration fixes exponent 0;
- the Remark 3.12.1(iii) arithmetic fact $\mathbb{Q}_{>0} \cap \mathbb{Z}^{\times} = \{1\}$;
- the Remark 3.12.1(ii) generalized-link numerical bound $C_\Theta \geq -\lambda$ for $\lambda \in \mathbb{Q}_{>0}$, with strict strengthening over the $\lambda = 1$ bound when $\lambda < 1$;
- the Remark 3.12.2 distinct-label implication q-itw $\Rightarrow$ q-itw $\wedge$ Θ-itw/indets, with q-native and Θ-native labels proved distinct;
- the Remark 3.12.2 toy calculation: forgotten concrete assignments $-h$ and $-2h$ are incompatible for $h > 0$, while $-h \in \mathbb{R}_{\leq -2h+\epsilon}$ implies $h \leq \epsilon$;
- the Fig. 3.9 column table: both $\bullet_0$ and $\circ$ roles are required in both columns, while q-pilot multiradiality is absent;
- the Remark 3.12.2(v) absorption distinction: zero-column unit indeterminacy is absorbed by a hull upper ray, while one-column logarithmic conjugacy is absorbed as log-volume equality;
- comparison levels: pointwise representative, aggregate representative, balanced sign-compatible, and hull/log-volume;
- endpoint audits for the 3.11.5 $\Rightarrow$ 3.12 comparison chart;
- nonarchimedean and archimedean (Ind3) local orientations;
- ordered real-line alignment with cancellation only after scale equality;
- a canonical unit-scale ordered alignment from a nonarchimedean (Ind3) entry;
- a bridge from factored square/full-label SHE obligations to the Hodge-descent (Ind3) route.

5 Current Lean Results

The current first-pass experiment report evaluates to:

route theorem flags	$=$ `true`,
collapse-obstruction flags	$=$ `true`,
Gaussian/procession finite-model flags	$=$ `true`,
`balancedLevelRejectedAtFinalRouteTheoremAvailable`	$=$ `true`,
`disputeSettledByCurrentStage`	$=$ `false`.

Here the route flags are ordered real-line, nonarchimedean entry, and factored SHE availability. The obstruction flags are raw-cancellation failure, label-independent j^2 failure, and sign-quotient representative descent failure. The Gaussian/procession flags include finite tensor-packet, monoid action, base-valuation product, coric-transfer, determinant-normalization, upper-ray, q-pilot two-computation, tensor-power warning, and global Frobenioid-calibration shadows, plus positive-rational unit rigidity. The generalized-link flag records the rational parameter inequality $C_\Theta \geq -\lambda$. The distinct-label flag records the logical intertwining display in Remark 3.12.2 and rejects collapse of the q-native and Θ-native labels. The toy-model flag records the resulting upper-ray arithmetic $h \leq \epsilon$. The column-table flag records the Fig. 3.9 theta/q similarity-difference split. The absorption flag records the zero-column upper-bound versus one-column equality distinction. The theorem-level route is:

$$\begin{array}{ccccc}
\text{nonarch. (Ind3) entry} & \rightarrow & \text{canonical ordered } \mathbb{R}\text{-alignment} & \rightarrow & \text{source/target alignment} \\
 & & \downarrow & & \downarrow \\
\text{factored SHE transport} & \rightarrow & \text{Hodge-descent audit} & \rightarrow & q_{\text{signed}} \leq \Theta_{\text{signed}}.
\end{array}$$

The route remains conditional on the displayed hypotheses.

6 Audit Findings

No formal statement currently proves IUT III, Corollary 3.12 from Mochizuki's published hypotheses. The strongest positive result is conditional: if the nonarchimedean (Ind3) entry, ordered real-line alignment, and Hodge/SHE square-weight transport audit are supplied, then Lean checks the final signed q-to-Θ inequality.

The strongest negative result is also limited: in the current $\mathbb{F}_\ell$-label model, Lean rejects a single label-independent scale accounting for all representative j^2 factors, and it rejects balanced sign-compatible evidence as the final hull/log-volume comparison level. Lean also rejects direct descent of arbitrary raw representative j^2 theta-degree data to $|F_\ell|$, while accepting the half-range exponent convention stated in IUT II. Lean also checks that an N-th tensor-power boundary is a strict strengthening for $N \geq 2$ and negative pilot log-volume. It also checks the Step (xii) numerical warning: nonzero local Frobenioid shifts change log-volume, whereas global calibration restores the unshifted value. The positive rational unit rigidity fact in Remark 3.12.1(iii) is formalized directly over $\mathbb{Q}$. For generalized links, Lean records that $\lambda < 1$ makes $-\lambda > -1$. Lean also checks that the Remark 3.12.2 simultaneous-intertwining step depends on distinct labels rather than an unlabeled identification. In the accompanying toy model, Lean proves the final upper bound exactly from the upper-ray membership hypothesis. Lean also records that the q-pilot column lacks the theta-side multiradiality used in Theorem 3.11(iii)(c). For Remark 3.12.2(v), Lean separates hull absorption of unit shifts from log-volume equality after logarithmic-conjugate compatibility.

7 Missing Mathematics

The present code does not yet formalize:

- initial Θ-data as in IUT I;
- actual Hodge theaters, Frobenioids, prime strips, log-shells, or Θ-links;
- the Hodge–Arakelov theta evaluation and Gaussian monoids of IUT II;

- the construction of the actual theta values, Gaussian/splitting monoids, $F^{\times}_{\mathrm{mod}}$-actions, Kummer isomorphisms, mono-analyticization, and Frobenioids behind the present shadows;
- the actual construction of log-shells and tensor packets attached to mono-analytic prime-strips in IUT III, Theorem 3.11;
- the genuine holomorphic hull and determinant operations of Step (xi), beyond the present finite determinant/upper-ray/two-computation/tensor-power shadows;
- the global realified Frobenioids of Step (xii), beyond the present integer log-volume shift model;
- the exact IUT IV log-volume estimates and Diophantine applications;
- a proof that the current Hodge/SHE transport obligations follow from Mochizuki's full hypotheses.

8 Next Work

The next significant formal step is not another wrapper. It is to replace current explicit Hodge/SHE transport assumptions by constructions closer to the papers:

$$\begin{gathered}\text{theta values } q^{j^2}\\ \rightsquigarrow \text{ Gaussian and splitting monoids}\\ \rightsquigarrow \text{ square/full-label preservation}\\ \rightsquigarrow \text{ hull/log-volume route.}\end{gathered}$$

Only after this bridge is formalized can the first-pass experiment begin to test the actual mathematical load-bearing part of Mochizuki's response to Scholze–Stix.

References

[1] S. Mochizuki, *Inter-universal Teichmuller Theory I: Construction of Hodge Theaters.*

[2] S. Mochizuki, *Inter-universal Teichmuller Theory II: Hodge-Arakelov-theoretic Evaluation.*

[3] S. Mochizuki, *Inter-universal Teichmuller Theory III: Canonical Splittings of the Log-theta-lattice.*

[4] S. Mochizuki, *Inter-universal Teichmuller Theory IV: Log-volume Computations and Set-theoretic Foundations.*

[5] S. Mochizuki, *On the Formalization of IUT: Preliminary Progress Report*, 2026.

[6] P. Scholze and J. Stix, *Why abc is still a conjecture.*